

Aradhya Yadav

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Professional Summary

AI/ML Engineer with end-to-end machine learning and deep learning project experience, specializing in Python, TensorFlow, Scikit-learn, and Large Language Models (LLMs). Designed and deployed three production-grade AI systems: an IoT anomaly detection model (94% detection accuracy, <2s latency), a deep ANN for molecular classification (88% validation accuracy, +50% reliability), and a multi-agent LLM policy forecasting pipeline (6 domain-specific agents analyzing 1,600+ speeches with zero data leakage). Hands-on with NLP, Transformers, Generative AI, RAG pipelines, feature engineering, and model evaluation. AWS-certified in ML, GenAI, and Data Analytics. Eager to contribute to production AI systems at scale.

Education

Master of Computer Applications (AI & ML) (2024–2026)

Chandigarh University, Gharuan

CGPA: **7.86**

Bachelor of Computer Applications (2021–2024)

Galgotias University, Greater Noida

CGPA: **8.62**

Intermediate (CBSE) (2019–2020)

Kendriya Vidyalaya No. 2, Armapur, Kanpur

Percentage: **71.0%**

Experience & Projects

Multi-Agent Mixture-of-Experts (MoE) Policy Forecasting Engine | May 2026

- **Architected** an AI forecasting system utilizing **6 specialized LLM agents** (Inflation, Jobs, GDP Growth, Financial Markets) powered by **LLaMA 3.3** via **Groq API** and **Pydantic** to analyze Federal Reserve speeches and predict rate decisions (CUT, HOLD, HIKE).
- **Built** a semantic retrieval pipeline using **Sentence Transformers (all-MiniLM-L6-v2)** to query relevant evidence across **1,600+ historical Fed speeches**, enforcing strict date filters to ensure zero lookahead data leakage.
- **Trained** classification models benchmarked against **Random Forest** and **XGBoost**, and deployed an interactive **Streamlit web dashboard** to visualize agent reasoning and real-time meeting forecasts.
- Deployed on Streamlit: <https://macro-policy-dashboard-9wr9yt3ei4qndv8kumvxzt.streamlit.app/>

AI-Based Early Anomaly Detection System for Energy & Server Infrastructure | Feb 2026

- Designed and deployed a real-time **sensor data pipeline** using **ACS712**, **DHT11**, and **MQ2 sensors** for capturing electrical, environmental, and gas concentration data, generating a custom **dataset of 5,000+ data points** for anomaly detection model training.
- Implemented unsupervised anomaly detection using **Isolation Forest algorithm**, achieving a **94% detection rate of simulated faults** across power usage and temperature behavior to enhance server infrastructure reliability.
- Integrated **ESP8266 for WiFi connectivity** and engineered an automated real-time alert system delivering anomaly notifications via email through the **Blynk IoT platform**, maintaining system latency under **2 seconds**.
- Github link : <https://github.com/Aradhya291/Early-Anamoly-detection>

Teaching Taste to AI using Deep Learning | Nov 2025

- Developed a **deep learning-based** predictive system to classify molecular taste profiles from chemical structures using **Artificial Neural Networks (ANN)** and advanced feature engineering techniques.
- Constructed a structured molecular feature database using **RDKit descriptors** and transformed raw chemical data into **ANN-ready predictive inputs** through systematic data preprocessing and feature extraction.
- Enhanced classification reliability by **50% over 4,000+ molecular structures** by redesigning **hidden layer architecture** and optimizing the input **feature engineering pipeline**.
- Trained and evaluated an **ANN-based** predictive model for molecular taste classification, achieving **88% validation accuracy** and **93% training accuracy**.
- Github link : <https://github.com/Aradhya291/Teaching-Taste-to-AI-using-Deep-Learning>

Certifications

- AWS Academy Graduate- [Generative AI Foundations](#) (Feb 2026)
- AWS Academy Graduate- [Machine Learning](#) (Oct 2025)
- DeepLearning.AI – [Sequence Models](#) (Oct 2025)
- Intel AI Impact Global Festival – [Generative AI](#) (Oct 2024)
- AWS Academy Graduate – [Data Analytics](#) (May 2023)

Technical Skills

Programming Language: Python

ML/DL Frameworks: TensorFlow, Keras, Scikit-learn

Libraries: Pandas, NumPy, Matplotlib, NLTK, OpenCV, RDKit

Software Tools: VS Code, Google Colab, Antigravity IDE, Arduino IDE

Database Management: MySQL

Concepts: Machine Learning, Deep Learning, Transformers, NLP, Generative AI , Retrieval Augmented Generation(RAG). Feature Engineering

Awards & Co-curricular Achievements

- **First Position** – UIC Chess Championship (Jan 2024)
- **Golden Volunteer** – Nari Shakti NGO (Jul 2022)
- **Participated in 50th KVS National Chess Meet 2019 (Under-17)** (Jun 2019)

Additional Information

Interests & Hobbies- Travelling, Chess | **Languages Known-** English,Hindi